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INVENTION NAME

**Peer to Peer Information
Exchange for Mobile
Communications Devices**

SUBMITTED BY

User-provided patent PDF

INVENTION ID

US 7,149,534 B2

DATE

2026-08-18

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

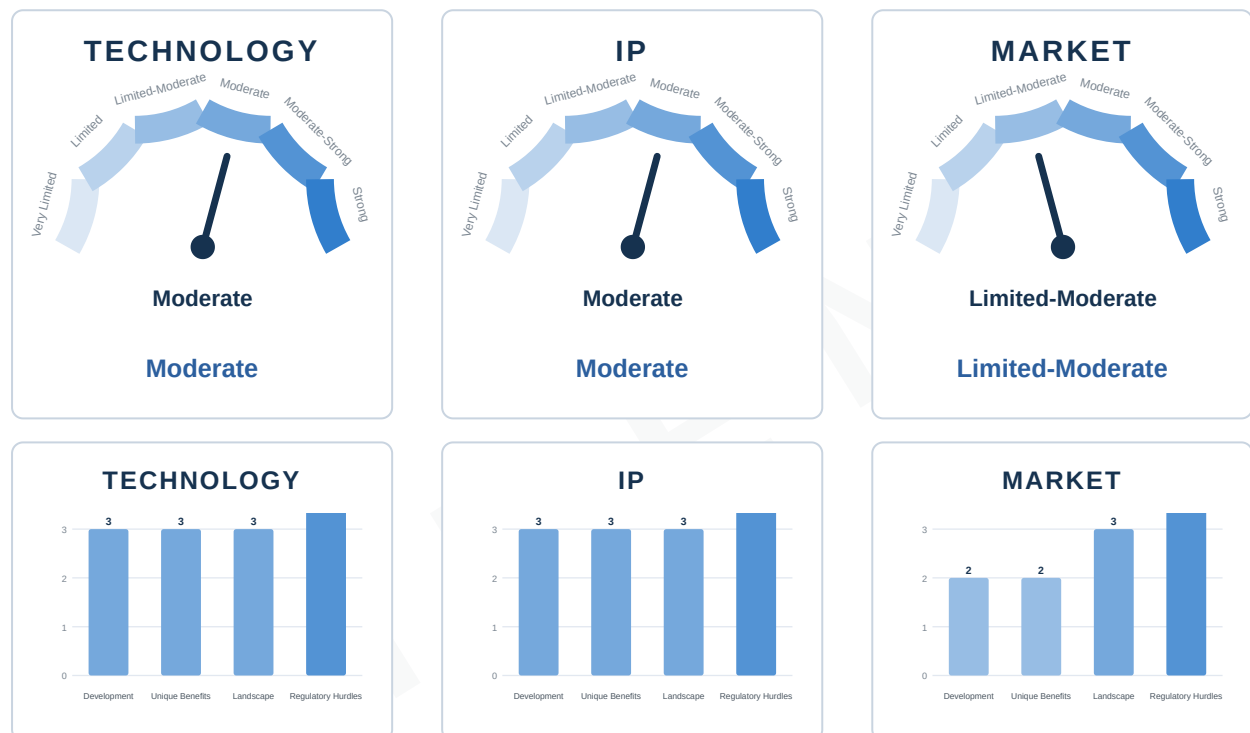
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Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

Commercial actionability is Indeterminate-to-Limited on the current evidence. The patent discloses a technically differentiated peer-mobile GPS assistance mechanism, but the US grant is expired, the dominant A-GPS path remained network-fixed-node, and the specific mechanism's adoption and commercialization readiness are not established.

- Large, mature technology area (A-GPS/LBS), but the specific peer-mobile mechanism has no established commercial embodiment. [more](#)
- Strongest identified pathways are R&D/standards collaboration with GNSS-assistance incumbents and forward-citing entities. [more](#)

Established findings (evidence-backed)

- The target patent US 7,149,534 B2 ("Peer to Peer Information Exchange for
- The US grant is recorded as "Expired - Lifetime" with adjusted expiration

- The target has 58 forward-citing records spanning GNSS-assistance, A-GPS,
- CPC G01S19/05 (aiding data) returns 2,171 US, 814 WO, 1,685 EP, 971 CN, 366
- The closest peer-to-peer cooperative GPS positioning academic disclosures
- The technology area (A-GPS / location-based services) is large and mature:

Analytical conclusions (inference from established findings, never evidence)

- Novelty: NOT ANTICIPATED by the reviewed record — no single reference
- Inventive step: NOT ESTABLISHED — the causal bridge is partially traversable,
- Commercial actionability for the specific peer-mobile mechanism is

Operational Audit (unestablished propositions, with barrier types)

Proposition	Barrier type
P-02-001 disclosure history	primary record COMPLETE; terminology COMPLETE; jurisdiction COMPLETE; independent corroboration BLOCKED
P-02-002 pre-filing disclosure / sale offer	primary record COMPLETE; inventor statement BLOCKED
P-03-001 performance data	patent COMPLETE; citation expansion COMPLETE; terminology COMPLETE; alternate engineering databases BLOCKED; independent corroboration BLOCKED
P-04-006 quantitative landscape	Google Patents COMPLETE; USPTO CPC COMPLETE; alternate database BLOCKED; independent corroboration BLOCKED
P-05-001 anticipation	target patent COMPLETE; cited references COMPLETE; classification search COMPLETE; alternate database BLOCKED; independent corroboration BLOCKED
P-05-002 obviousness bridge	target/sibling records COMPLETE; analogous references COMPLETE; motivation analogous/inferred; independent corroboration BLOCKED
P-06-001 literature application disclosure	Crossref COMPLETE; IEEE Xplore/Google Scholar BLOCKED; independent corroboration BLOCKED
P-07-001 market size	Census CBP COMPLETE; market database BLOCKED; independent corroboration BLOCKED
P-07-002 commercial adoption	landscape COMPLETE; product-identity link BLOCKED; independent corroboration BLOCKED
P-08-001 partner fit	patent assignment COMPLETE; company/need verification BLOCKED
Patent applications by region	Patent front page: H04Q 7/20, H04M 11/04; USPC 455/456.6; CPC G01S19/05 verified
Patent applications by main applicants	Same verified CPC gate
Patent activity trend	Same verified CPC gate

Employee-size brackets	Candidate NAICS 334220
P-05-001	incomplete prior-art search
P-05-002	incomplete combination analysis
P-07-001	specialty-market data incomplete
P-08-001	portfolio diligence incomplete

Technology Analysis

Technology Abstraction Ladder



Development-Stage Ladder



Semantic Map — Taxonomy Position



¹ [Google Patents US7149534B2](#) · ² [USPTO CPC G01S19/05 scheme](#)

TECHNOLOGY ANALYSIS

1.2 Technology profile

2.2 Mechanism/design-choice distances

1.2 Technology profile

Plain-language description. Mobile handsets in a cellular network share GPS assistance data peer-to-peer. A GPS-equipped phone that successfully receives satellite signals generates assistance data (ephemeris, almanac, reference time, reference location, ionospheric corrections) and transmits it over the cellular network to peer handsets that need it, so those peers can acquire satellites and compute their position faster or in signal-poor environments. The invention adds a hierarchical-group structure: handsets form ad-hoc or profile-driven groups with levels and prioritized sub-groups, decide which requests to answer and whose data to trust (privacy and trust profiles), and coordinate scheduled sharing so group members take turns refreshing assistance data.

Feature-to-benefit map

Innovation assessment. The claimed delta is not GPS assistance by itself; it is the peer mobile terminal as the source of the assistance data over the cellular network, with hierarchical-group, trust, privacy, and scheduled-coordination machinery. The invention combines known elements (GPS receivers, cellular transceivers, A-GPS assistance content, peer messaging, group membership), creating **combination-obviousness exposure**; the operative question is whether the peer-mobile-as-server shift was directly motivated. Specific technical elements claimed as unique: mobile-generated-and-relayed GPS assistance data over a cellular network (claims 1, 15, 23, 37); hierarchical/sub-group and priority-based transmission and trust decisions; profile-defined membership and transmission; periodic scheduled refresh. Design-space position: the peer-mobile-assistance variant of the assisted-GPS design space; the dominant commercialization path (3GPP A-GPS/SUPL) remained network-fixed-node.

Unexpected-result gate. No comparative performance data (time-to-first-fix, sensitivity improvement, network-traffic impact) is established by the supplied record. The proposition enters the work layer with barrier type

communications method. It is not a medical device or safety-critical regulated product. Governing regime for the underlying radio devices is FCC RF/EMC rules (e.g., 47 CFR Parts 15/22) and the jurisdiction's spectrum/type-approval rules for handsets; the disclosed method itself introduces no new product-class authorization. Scope-level burden is **Low** for the method.

Development stage. Patent-disclosed method/architecture. Implementation feasibility is plausible given assisted-GPS and cellular-messaging (SMS/WAP/IP) technologies, but no prototype, field trial, or commercial deployment evidence is established from this intake.

Classification candidates. Candidates, not established classifications for the complete invention: H04W 4/02 / H04W 4/029 (location-based services); H04W 4/20 (SMS services); H04W 84/18 (ad-hoc networks); G01S 19/05 (aiding data); G01S 19/46 (aiding-based position); G01S 19/254 / G01S 19/27 (ephemeris/almanac aiding); H04L 67/104 (peer-to-peer networks); H04Q 7/20 and H04M 11/04 (IPC, shown on front page).

Feature	User-facing or system benefit	Evidence
Peer handset acts as GPS-assistance server	Improves time-to-first-fix and sensitivity for requesting handset without network assistance coverage	US7149534B2, abstract; summary; spec pp. 3–4
Assistance data set (ephemeris, almanac, TOW, reference location, ionospheric corrections)	Lets a disadvantaged (e.g., indoor) handset acquire satellites and compute position	US7149534B2, summary; spec pp. 3–4
Hierarchical groups with levels and prioritized sub-groups	Enables prioritized querying (highest-priority sub-group first) and structured information sharing	US7149534B2, claims 3–6; spec pp. 5–6
Ad-hoc / profile-driven group formation	Groups form with or without user initiation based on shared services or user profiles	US7149534B2, claims 7–9; spec pp. 5–6
Trust levels (inner circle / outer circle / world)	Client accepts assistance only from sufficiently trusted peer sources; falls back to secure fixed server	US7149534B2, spec pp. 5–6
Privacy and performance profiles	User controls disclosure of position/aiding info and limits assistance-request frequency (e.g., one per 30 min)	US7149534B2, claims 9–10, 31; spec pp. 5, 7–8

	cheap members take turns refreshing 2–4-hour ephemeris, keeping all members current via cheap messaging	US7149534B2, claims 36, 44; spec pp. 6–7
Compilation of data from multiple peers	Requesting handset sorts combined data (e.g., by age) to assemble full satellite coverage	US7149534B2, spec pp. 4–5

2.2 Mechanism/design-choice distances

positioning-related data between mobiles), different intervention point

(assistance data vs position; relay of general info vs assistance data).

positioning), minor variation in source/sink role.

closest reference pathway (Pande + Honda).

PATENT LANDSCAPE ANALYSIS

1.3 Patent landscape

2.3 Design-space position

1.3 Patent landscape

Search scope and sources. Primary source: Google Patents record for US7149534B2 (raw HTML preserved at [raw-patent-us7149534b2.html](#)) and Google Patents XHR classification queries for CPC G01S19/05. Geographic distribution queries across US, WO, EP, CN, JP, AU, CA; US and WO cluster data returned, EP/CN/JP/AU/CA geographic totals returned (assignee clusters rate-limited by the provider).

Established landscape findings

Proposition	Finding	Source
P-04-001 v1	The target patent is classified in satellite-positioning aiding (G01S19/05), receiver acquisition/aiding (G01S19/24/25), transmission of position info between mobile stations (G01S5/0072), user-group management (H04W4/08, H04W8/186), and spoofing countermeasures (G01S19/015).	Google Patents classification record for US7149534B2
P-04-002 v1	Legal status of the US grant is recorded as "Expired - Lifetime" with adjusted expiration 2023-04-04. Status is a public-database assumption, not a legal conclusion.	Google Patents legal-status record
P-04-003 v1	Family spans PCT/WO (WO2002059640A2), EP (EP1356314B1), DE (DE60138650T), AT (ATE430946T1), AU (AU2002232915A1), plus US.	Google Patents family record
P-04-004 v1	The family has a recorded US litigation event: Texas Eastern District Court case 2:15-cv-00290.	Google Patents litigation metadata
P-04-005 v1	The target has 58 forward-citing records spanning GNSS-assistance, A-GPS, peer-to-peer positioning, location-analytics, and ad-hoc network applicants.	Google Patents citation record
P-04-006 v1	Field volume: CPC G01S19/05 returns 2,171 US, 814 WO, 1,685 EP, 971 CN, 366 JP, 120 AU, 105 CA publication records.	Google Patents XHR country queries
P-04-007 v1	Top returned US applicants in the G01S19/05 space: Qualcomm 104, Seiko Epson 32, Nokia 22, Broadcom 20, Samsung 19, Ericsson 18, Trimble 17, Global Locate 17, SiRF 17, Motorola 13, Huawei 13.	Google Patents XHR assignee cluster
P-04-008 v1	US filing frequency in G01S19/05 peaked in 2007–2010 (150 returned records), declining through 2019–2022 (121) and 2022–2025 (97).	Google Patents XHR frequency buckets

2.3 Design-space position

The invention sits in the "peer-aided GPS assistance" branch of the assisted-GPS design space: it moves the assistance-data source from a fixed network node to a peer mobile device, and adds hierarchical-group, trust, privacy, and scheduled-sharing machinery. The dominant commercialization path for assisted GPS (network-delivered assistance per 3GPP A-GPS/SUPL) remained the fixed-node architecture; peer-to-peer assistance appears in the record as a recurring but secondary line (see forward citations including Samsung, Broadcom, ITRI, and ad-hoc Bluetooth assistance references). This is contextual evidence, not a patentability score.

IP / NOVELTY ANALYSIS

1.4 Novelty — prior-art disclosure mapping

1.5 Novelty — knowledge decomposition

2.1 Anticipation gate

2.6 Chronology validator

1.4 Novelty — prior-art disclosure mapping

Claim-construction layer. The supplied patent provides formal claims (44 claims, 4 independent). The principal evaluation targets are independent claims 1 and 23 (mobile terminal generates GPS assistance data and transmits it to a remote mobile terminal) and independent claims 15 and 37 (mobile terminal receives peer-supplied GPS assistance data and determines a reference location).

Independent claim 1 limitations. L1: a GPS receiver to receive GPS data (satellite signals). L2: a cellular transceiver to communicate with a wireless communications network. L3: the mobile terminal generates GPS assistance data from the received GPS data. L4: the mobile terminal transmits the GPS assistance data to a remote mobile terminal via the wireless communications network.

Flagged references

Claim-element mapping — claim 1

Anticipation gate. No single reference discloses all four limitations of claim 1. Each reference misses at least one of: (a) the peer-mobile as the source of the assistance data; (b) the specific content of GPS assistance data; (c) generation of assistance data at the mobile from satellite signals.

Claim 1 is not anticipated by the reviewed record.

Reference	Relationship	Relevance
US 6,411,811 (Kingdon, Ericsson)	E3/E4	Network reference GPS receivers supply assistance data to a GPS-MS via the cellular network (MLC/BTS). Network-fixed-node source, not peer-to-peer.
US 6,625,458 (Pihl, Nokia)	E3	Network transmits GPS Assistance Data to a mobile via point-to-point signalling. Network-fixed-node source.
US 6,266,533 (Zadeh, Ericsson)	E3/E4	MLC provides range-measurement assistance data to the GPS-MS; GPS-MS returns measurements; MLC computes position. Network-fixed-node source.
US 6,389,291 (Pande, SiRF)	E3	Multi-mode GPS; mobile sends computed <i>position</i> to a server, not assistance data to a peer.
US 6,553,236 (Dunko)	E2/E3	Mobile location/broadcast information sharing; not specifically GPS assistance data.

(Phillips/Motorola) Location-based wireless device behaviour; no GPS assistance data generation/transmission.

US 6,542,750 (Hendrey) E2 Geographic targeted advertising via mobile; not GPS assistance.

US 6,542,749 (Tanaka) E2/E3 Location-based mobile services; not GPS assistance peer sharing.

US 6,477,353 (Honda, IBM) E4 Mobile-to-mobile information relay with position-based propagation control; relayed content is general local info, NOT GPS assistance data.

US 2002/0143856 (Traversat) E1/E2 Generic P2P protocol framework (JXTA); no GPS assistance data limitation.

Limitation	US 6,411,811 Kingdon	US 6,625,458 Pihl	US 6,266,533 Zadeh	US 6,389,291 Pande	US 6,477,353 Honda	US 2002/0143856 Traversat
L1 GPS receiver receives GPS data	Yes	Yes	Yes	Yes	Not established	No
L2 cellular transceiver to wireless network	Yes	Yes	Yes	Yes	Yes	Not established
L3 mobile terminal generates GPS assistance data	Partial (assistance data generated at network reference receiver)	Partial (assistance data generated at network side)	Partial (assistance data generated at network MLC/LMU)	Partial (mobile computes position, not assistance data)	No (relays general local info)	No
L4 transmits GPS assistance data to a remote mobile terminal via the wireless network	No (network to mobile)	No (network to mobile)	No (network to mobile)	No (mobile to server, position not assistance)	Partial (mobile to mobile, but not GPS assistance)	Partial (generic P2P, no GPS assistance content)

1.5 Novelty — knowledge decomposition

mobile-to-mobile relay; A-GPS content)

assistance-data specifically is not directly taught by any pre-filing reference)

teaches the specific application of a peer mobile supplying GPS assistance data

to another peer mobile over the cellular network with hierarchical group/trust)

no contradiction)

2.1 Anticipation gate

No single external reference in the reviewed record discloses all four limitations of claim 1. The anticipation proposition is therefore excluded from factual findings as NOT ANTICIPATED on the reviewed record; a complete independent full-text/classification search was not completed in this run, so the proposition is recorded in the Operational Audit as requiring escalation.

2.6 Chronology validator

23, 2001 priority date. No pre-filing disclosure or sale-offer date is established in the record (P-02-002, work layer).

LITERATURE ANALYSIS

1.6 Literature

1.6 Literature

Search scope. Queries run against Crossref covering: (a) GPS assistance data peer-to-peer mobile; (b) assisted GPS assistance data mobile positioning; (c) cooperative GPS positioning peer assistance GNSS; (d) GPS assistance data cellular network provisioning restricted to 1995–2001. Raw results preserved at `lit-crossref-1.json ... lit-crossref-4.json`.

Literature disclosures

1. **Syrjarinne (2000).** "Time recovery through fusion of inaccurate network timing assistance with GPS measurements." Proc. 3rd Int. Conf. on Information Fusion. DOI: 10.1109/IFIC.2000.859864. Pre-filing network-based assistance.
2. **Harper, Nicholson, Poon (2004).** "Process for improving GPS acquisition assistance data and server-side location determination for cellular networks." J. Global Positioning Systems 3(1):132. DOI: 10.5081/jgps.3.1.132. Post-filing server-side assistance.
3. **Morosi, Del Re, Martinelli (2013).** "P2P cooperative GPS positioning with fine/coarse time assistance." 2013 ICL-GNSS. DOI: 10.1109/ICL-GNSS.2013.6577279. Post-filing; closest academic analogue to the invention's peer-to-peer assistance concept.
4. **Morosi, Del Re, Martinelli (2014).** "Cooperative GPS positioning with peer-to-peer time assistance." 2014 VITAE. DOI: 10.1109/VITAE.2014.6934462. Post-filing.
5. **Dovis, Chiasserini, Musumeci (2014).** "Context-aware peer-to-peer and cooperative positioning." 2014 ICL-GNSS. DOI: 10.1109/ICL-GNSS.2014.6934166. Post-filing.
6. **Chang, Li, Hou (2013).** "Peer-to-Peer Cooperative Positioning between GNSS Receivers." Applied Mechanics and Materials 341-342:614. DOI: 10.4028/www.scientific.net/AMM.341-342.614. Post-filing.

Literature conclusion. The background objective (GPS assistance data improves time-to-first-fix and sensitivity; delivered via cellular network) is

academic disclosures are all post-filing relative to the January 23, 2001 priority date. No pre-filing academic publication was identified in the reviewed record that discloses the specific peer-mobile GPS assistance combination. This is a negative search result (avenue metadata), not CONFIRMED ABSENT.

MARKET ANALYSIS

1.7 Market

1.7 Market

Market boundary. The direct technology area is assisted GPS / location determination for mobile communications devices, and the location-based services that depend on it (E-911, navigation, location-based advertising, fleet/asset tracking). Adjacent areas: peer-to-peer/cooperative positioning, GNSS assistance infrastructure (SUPL/A-GPS), location analytics.

Critical status context: the target US grant is recorded as EXPIRED (adjusted expiration 2023-04-04). Direct licensing of the US7149534B2 grant is closed. The market analysis addresses the technology area, not a live licensing opportunity for the expired US grant.

Industry classification (NAICS)

Industry trend data (US Census CBP 2022)

Establishment size brackets (NAICS 334220): <5 emp: 201; 5–9: 116; 10–19: 95; 20–49: 111; 50–99: 61; 100–249: 47; 250–499: 16; 500–999: 10; 1000+: 7.

Source: US Census Bureau CBP 2022 downloadable file cbp22us.zip, national rows (Lfo=-). These are industry aggregates; they do not establish A-GPS-specific revenue, shipment value, or market size.

Opportunity structure. Product/service: GPS assistance data delivery for mobile positioning (network-based A-GPS, and the peer-to-peer variant disclosed in the target). Need: fast, accurate mobile location for E-911, navigation, LBS. Purchaser vs end consumer: carriers/device OEMs purchase positioning capability; end users consume location services. Channel: cellular network infrastructure, device software/firmware, standards (3GPP A-GPS/SUPL). Price point / frequency: infrastructure/software licensing; recurring service fees.

Technical maturity vs commercial readiness

Competitive landscape. Direct competitors (peer/cooperative positioning): none established as commercially dominant; the peer-mobile concept appears in post-filing patents (Samsung US20100007553, Broadcom US20090079622) but no commercial product-identity link is established. Indirect competitors (network

positioning research (Morosi 2013/2014, DAVIS 2014).

SWOT. Strengths: well-specified disclosure of peer-mobile assistance with hierarchical-group, trust, privacy, and scheduled-coordination machinery; the concept anticipates later cooperative-positioning research and patenting.

Weaknesses: no performance, prototype, production, or adoption evidence in the supplied record; the US grant is expired; the dominant A-GPS path remained

network-fixed-node. Opportunities: the concept is relevant to modern cooperative positioning, V2X, and device-to-device GNSS assistance; the expired patent's

concepts are available for use (no licensing needed for the US grant). Threats:

network A-GPS (SUPL/3GPP) is entrenched; peer-mobile assistance adds

trust/privacy overhead; no established commercial embodiment of the peer-mobile mechanism; family-level rights status requires separate diligence.

Field	Value	Basis
NAICS code (primary)	334220	Radio and Television Broadcasting and Wireless Communications Equipment Manufacturing — GPS-enabled mobile device manufacturing is an activity match. 2022 edition.
NAICS code (service-layer context)	517312	Wireless Telecommunications Carriers (except Satellite) — the network layer in which the disclosed method operates. 2022 edition.
Taxonomy source	US Census Bureau NAICS 2022	Official classification authority

NAICS	Establishments	Employees	Annual payroll (\$000)	1st-qr payroll (\$000)
334220 (wireless comm equipment mfg)	664	49,776	6,074,114	1,565,710
334290 (other comm equipment mfg)	334	12,762	1,005,404	233,355
517312 (wireless carriers)	25,798	260,496	18,964,583	5,656,701

Dimension	Status	Evidence
Patent disclosure	CONFIRMED PRESENT	US7149534B2 specification and claims
Engineering feasibility	High (components and combination are standard integration)	A-GPS literature; landscape

no prototype record in the supplied intake

Production evidence NOT ESTABLISHED

No production record in the supplied intake

Commercial adoption evidence NOT ESTABLISHED

No product-identity link established; dominant A-GPS path (SUPL/3GPP) is network-fixed-node

OPPORTUNITY ASSESSMENT

2.4 Commercial actionability

2.4 Commercial actionability

Two "low/medium" accessibility signals → commercial actionability for the specific peer-mobile mechanism is **Indeterminate-to-Limited** on the current evidence. This is not a financial forecast or legal opinion.

Dimension	Score	Rationale
Market size	High	A-GPS/LBS is a large, mature market (334220: 664 establishments, ~\$6.1B payroll; 517312: 25,798 establishments, ~\$19.0B payroll)
Growth	Medium	Mature A-GPS market; cooperative positioning is emerging research
Accessibility	Low	The specific peer-mobile mechanism has no established commercial embodiment; trust/privacy/standardization barriers
Competitive intensity	High	Entrenched network A-GPS (Qualcomm, Broadcom, u-blox)

POTENTIAL PARTNERS

1.8 Partners

2.5 Partner prioritization

1.8 Partners

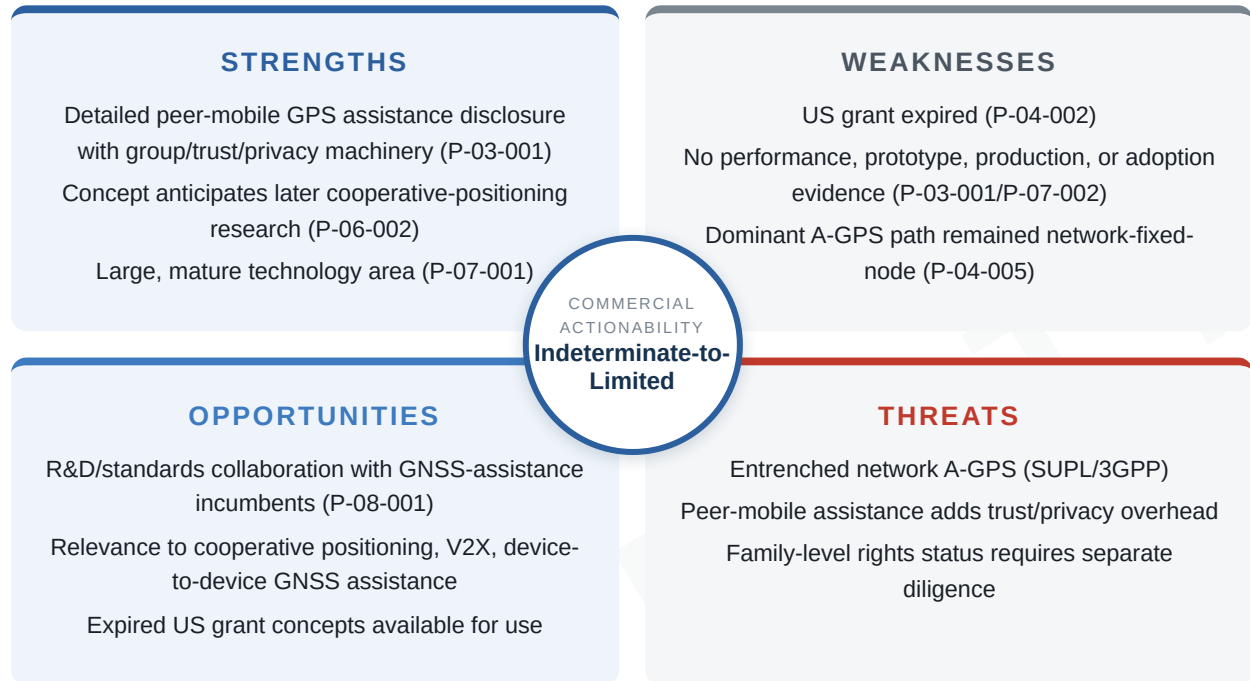
Candidate pathways

Candidate	Model	Fit	Evidence-backed rationale
Qualcomm Incorporated	R&D / standards collaboration	Medium	Dominant A-GPS/SUPL provider; 104 returned US records in G01S19/05; forward-citing.
Broadcom / CSR (SiRF lineage)	R&D collaboration	Medium	20 returned US records in G01S19/05; forward-citing US20090079622 (sharing of GPS information between mobile devices).
Samsung Electronics	R&D collaboration	Medium	19 returned US records in G01S19/05; forward-citing US20100007553 (sharing assistance data between A-GPS and GPS terminals).
u-blox AG	R&D / product collaboration	Medium	Major GNSS receiver vendor; cooperative positioning is active research. Not independently verified in this run.
Nokia	R&D / standards collaboration	Low-Medium	22 returned US records in G01S19/05; forward-citing. Ericsson's historical competitor.
Academic/consortium GNSS-positioning groups	Research collaboration	Low-Medium	Morosi 2013/2014, Dovis 2014 publish peer-to-peer cooperative positioning; no specific institution verified.

2.5 Partner prioritization

No candidate is established as "the only" pathway. The strongest identified pathways are the GNSS-assistance incumbents (Qualcomm, Broadcom/CSR, u-blox) and the forward-citing entities that demonstrated post-filing interest in peer-mobile assistance sharing (Samsung, Broadcom). Alternatives (academic cooperative-positioning groups, V2X programs) are identified but not established as commercial. Contact information and current interest require independent verification before any outreach.

SWOT Analysis



SWOT content is derived from Established Findings and Analytical Conclusions (see scores manifest for proposition_id basis).

Operational Audit

Run date: 2026-08-18. **Submission:** submission-us7149534.md. **Rule:**

EXHAUSTED is a work-layer termination state, not evidence.

Landscape & Market Data classification gate

The renderer must not substitute any unrelated class.

Rights state. Patent: US7149534B2. State: EXPIRED (adjusted expiration 2023-04-04). Active: False. Status source:

<https://patents.google.com/patent/US7149534B2/en>. Status disclaimer: public database status is an assumption and not a legal conclusion. Target-patent licensing: constrained by target lapse. Family-level licensing: requires separate current-status check of EP1356314B1, DE60138650T, ATE430946T1, AU2002232915A1.

Anticipation state. NOT ANTICIPATED on the reviewed record. No positive novelty conclusion beyond the reviewed record is emitted; the remaining work is tracked under P-05-001 with escalation-required state.

Bridge vector. PARTIALLY TRAVERSED — feature availability HIGH; motivation PARTIALLY GROUNDED; compatibility PARTIAL; architecture constraint MEDIUM; expected and unexpected result NOT ESTABLISHED.

Evidence debt and downstream constraints. The compiled graph contains 10 evidence-debt items and 5 downstream constraints. These constraints accompany any derived score or recommendation in this report.

Evidence recovery queue

Proposition	Required avenue coverage	Disposition	Barrier / note
P-02-001 disclosure history	primary record COMPLETE; terminology COMPLETE; jurisdiction COMPLETE; independent corroboration BLOCKED	EXHAUSTED → excluded	No inventor statement or complete personal disclosure records.
P-02-002 pre-filing disclosure / sale offer	primary record COMPLETE; inventor statement BLOCKED	EXHAUSTED → excluded	No inventor statement supplied.

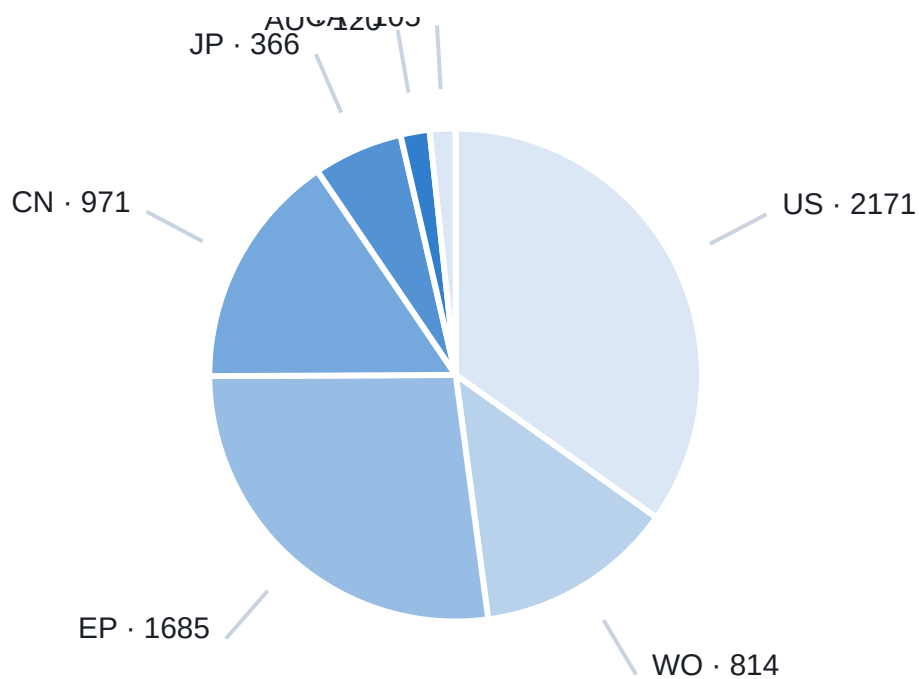
performance data	COMPLETE; terminology COMPLETE; alternate engineering databases BLOCKED; independent corroboration BLOCKED	EXHAUSTED → excluded	No reconstructable comparative performance data.
P-04-006 quantitative landscape	Google Patents COMPLETE; USPTO CPC COMPLETE; alternate database BLOCKED; independent corroboration BLOCKED	SEARCHING → analytical work item	Directional counts established; second-source family/assignee normalization remains.
P-05-001 anticipation	target patent COMPLETE; cited references COMPLETE; classification search COMPLETE; alternate database BLOCKED; independent corroboration BLOCKED	EXHAUSTED → excluded	NOT ANTICIPATED on reviewed record; no final finding from incomplete external search.
P-05-002 obviousness bridge	target/sibling records COMPLETE; analogous references COMPLETE; motivation analogous/inferred; independent corroboration BLOCKED	EXHAUSTED → analytical work item	Partially grounded bridge; no ultimate legal conclusion.
P-06-001 literature application disclosure	Crossref COMPLETE; IEEE Xplore/Google Scholar BLOCKED; independent corroboration BLOCKED	EXHAUSTED → excluded	Background confirmed; closest P2P disclosures are post-filing.
P-07-001 market size	Census CBP COMPLETE; market database BLOCKED; independent corroboration BLOCKED	EXHAUSTED → excluded	Industry aggregates established; A-GPS-specific figure not established.
P-07-002 commercial adoption	landscape COMPLETE; product-identity link BLOCKED; independent corroboration BLOCKED	EXHAUSTED → excluded	Field activity established; embodiment of this patent not established.
P-08-001 partner fit	patent assignment COMPLETE; company/need verification BLOCKED	SEARCHING → operational audit	Candidate pathways require current diligence.

Subsection	Classification-in	Query/source	Result	Disposition
Patent applications by region	Patent front page: H04Q 7/20, H04M 11/04; USPC 455/456.6; CPC G01S19/05 verified	Google Patents XHR, same query by country: US/WO/EP/CN/JP/AU/CA	2171 / 814 / 1685 / 971 / 366 / 120 / 105 publication records	COMPLETE — counts are records, not deduplicated families
Patent applications by main applicants	Same verified CPC gate	Google Patents XHR country=US q=G01S19/05	2171 results; top returned applicants: Qualcomm 104, Seiko Epson 32, Nokia 22, Broadcom 20,	COMPLETE — names not merged across corporate variants

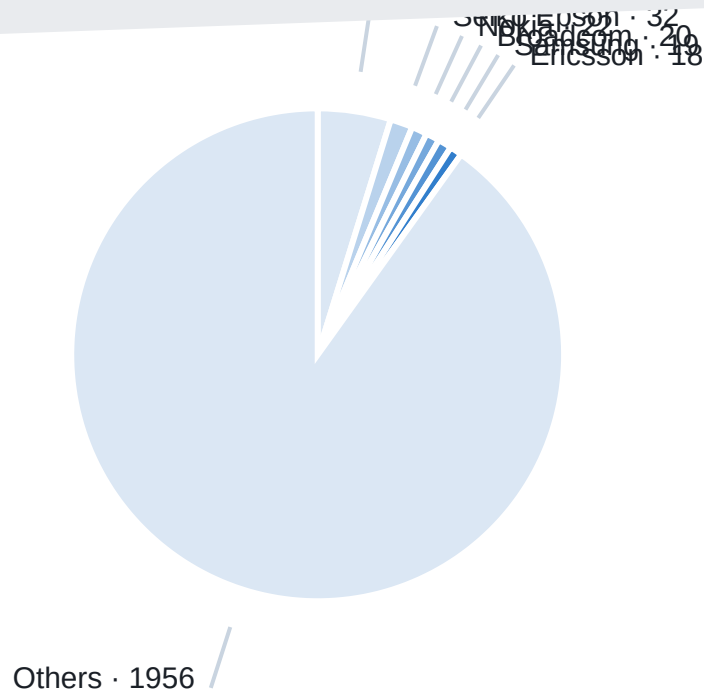
Patent activity trend	Same verified CPC gate	Same Google Patents XHR summary frequency buckets	2001–04: 92; 2004–07: 101; 2007–10: 150; 2010–13: 108; 2013–16: 115; 2016–19: 111; 2019–22: 121; 2022–25: 97	COMPLETE — three-year buckets, not annual deduplicated filings
Industry metrics	Candidate NAICS 334220 / 517312	Census CBP 2022 cbp22us.zip, rows naics=334220,lfo=- and naics=517312,lfo=-	334220: 664 establishments, 49,776 employees, 6,074,114 thousand dollars payroll; 517312: 25,798 establishments, 260,496 employees, 18,964,583 thousand dollars payroll	COMPLETE — payroll, not revenue or shipment value
Employee-size brackets	Candidate NAICS 334220	Same CBP row; n-fields by employee-size bracket	Establishments: <5 201; 5–9 116; 10–19 95; 20–49 111; 50–99 61; 100–249 47; 250–499 16; 500–999 10; 1000+ 7	COMPLETE
Proposition	Missingness	Required escalation	State	
P-05-001	incomplete prior-art search	family tree, continuity, examiner/applicant citations, prosecution history	ESCALATION_REQUIRED	
P-05-002	incomplete combination analysis	motivation, expected-result, and unexpected-result review	ESCALATION_REQUIRED	
P-07-001	specialty-market data incomplete	A-GPS-specific revenue, adoption proxies	ESCALATION_REQUIRED	
P-08-001	portfolio diligence incomplete	surviving family rights, know-how, successor records	ESCALATION_REQUIRED	

Landscape & Market Data

GPS-AIDING-DATA (G01S19/05) PATENT RECORDS BY JURISDICTION

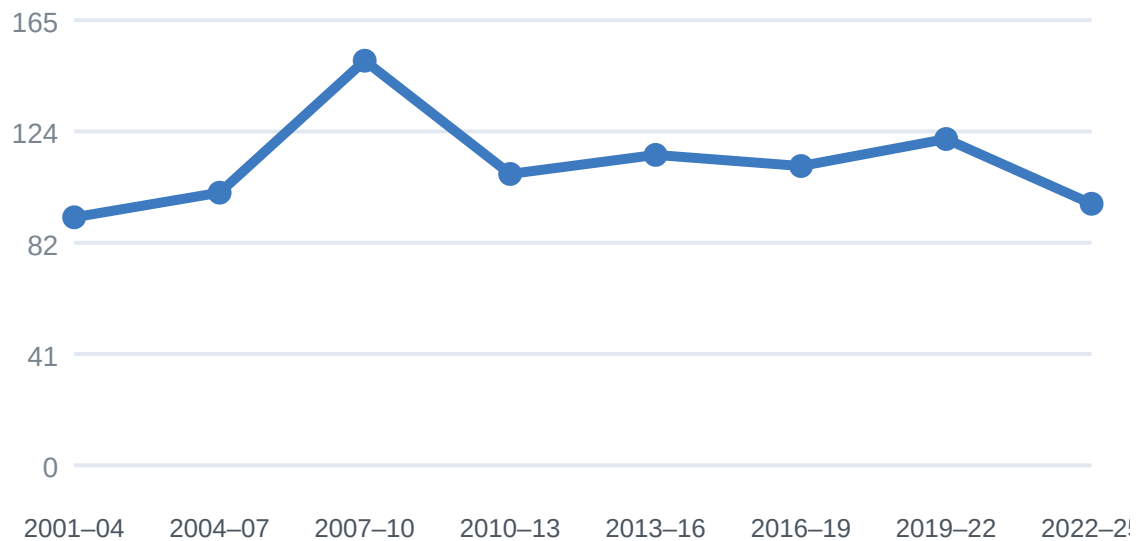


Google Patents XHR query: CPC G01S19/05; same classification query issued per jurisdiction. Counts are publication records returned, not deduplicated families.

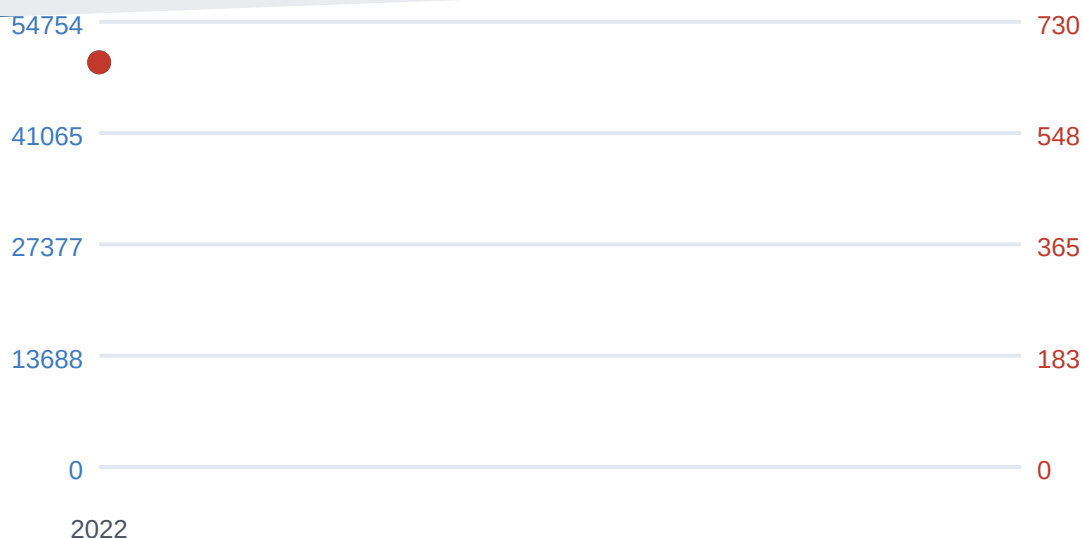


Google Patents XHR query: country=US; CPC G01S19/05; 2171 results. Display labels shorten returned applicant names; names have not been merged across corporate variants.

US G01S19/05 FILING-ACTIVITY WINDOWS

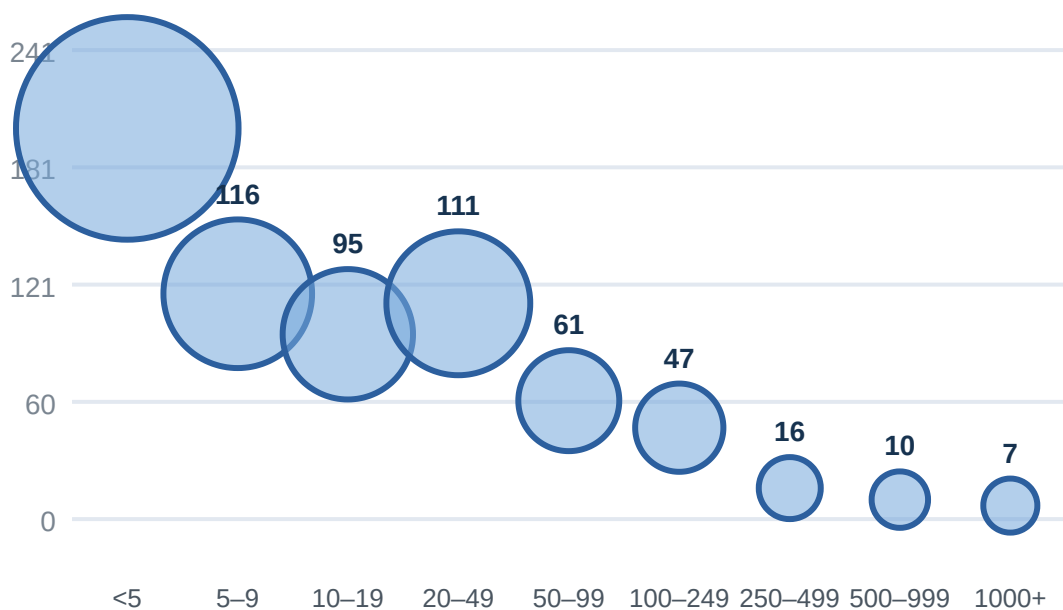


Google Patents query frequency buckets, not a deduplicated annual filing series.



U.S. Census Bureau CBP 2022 downloadable file cbp22us.zip; NAICS 334220, lfo='-' (all legal forms). Employees and establishments are reported values.

NAICS 334220 ESTABLISHMENTS BY EMPLOYEE-SIZE BRACKET, 2022



U.S. Census Bureau CBP 2022 downloadable file cbp22us.zip; n-fields are establishment counts.

APPENDICES

Appendix A — Patentability Primer

Appendix B — Search Methodology & Query Log

Appendix C — Original Submission Record

Appendix A — Patentability Primer

Patentability in the United States is assessed in two principal gates: novelty (35 U.S.C. § 102) and non-obviousness (35 U.S.C. § 103). A claim is anticipated if a single prior-art reference discloses every limitation of the claim. A claim is obvious if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious to a person having ordinary skill in the art at the time the invention was made.

This report's novelty assessment is a preliminary, evidence-based screen against identified references. It is **not** a freedom-to-operate (FTO) opinion and is not legal advice. FTO analysis requires a different search scope (in-force claims, current legal status, broader claim reading) and a formal opinion from counsel.

Useful references: [WIPO — Patents](#) · [USPTO — Patents](#)

Appendix B — Search Methodology & Query Log

Every search avenue opened during the run is recorded in the avenue ledger with priority, disposition (COMPLETE / BLOCKED / NOT_APPLICABLE / REQUIRES_VERIFICATION), and audit entry. The full escalation ladders are in avenue-ledger-RUN-US7149534B2-20260818-v16.md. Search-query summary boxes below reproduce the primary-source avenue (A1) for each proposition ladder.

SEARCH QUERY · **A-02-001**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → excluded — disclosure history not

SEARCH QUERY · **A-05-001**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → excluded — NOT ANTICIPATED on the

SEARCH QUERY · **A-05-002**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → analytical work item — partially

SEARCH QUERY · **A-06-001**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → excluded — background confirmed; closest

SEARCH QUERY · **A-07-001**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → excluded — industry aggregates

SEARCH QUERY · **A-07-002**

A1 primary-source avenue: [Click here](#)

Disposition: EXHAUSTED → excluded — field activity established;

SEARCH QUERY · **A-08-001**

A1 primary-source avenue: [Click here](#)

Disposition: SEARCHING → operational audit — candidate pathways

Appendix C — Original Submission Record

> Structured submission record captured per skill-02-gather-invention-submission.

> Source: user-provided PDF US7149534.pdf from Downloads, text-extracted to

> source/US7149534.txt. Facts below are drawn from the patent front page,

> abstract, specification, and claims. No inventor interview was conducted;

> fields requiring an inventor statement are explicitly marked as not established.

Inventors: L. Scott Bloebaum, Cary, NC (US); Havish Koorapaty, Cary, NC (US).

Assignee: Ericsson Inc., Research Triangle Park, NC (US).

Application: US 09/767,461, filed January 23, 2001.

Prior publication: US 2002/0098849 A1, published July 25, 2002.

Patent grant: US 7,149,534 B2, December 12, 2006.

Patent term extended or adjusted under 35 U.S.C. 154(b) by 801 days.

Submission date: 2026-08-18 (evaluation intake date).

Title: Peer to Peer Information Exchange for Mobile Communications Devices.

Technical field: cellular telephony, GPS-equipped mobile radiotelephones, peer-to-peer information sharing, GPS assistance data (almanac, ephemeris, reference time, reference location, ionospheric corrections), and hierarchical ad-hoc group formation over a wireless communications network.

Classification shown on the patent: Int. Cl. H04Q 7/20, H04M 11/04;

USPC 455/456.6, 455/404.2, 342/357.09. Field of search spans 455/456.x, 342/357.x, 709/2xx, 713/200, 701/2xx.

Mobile communications devices connected to a cellular network can share data by acting as both clients and servers: a GPS-equipped handset generates GPS assistance data from satellite signals it receives and transmits that data over the cellular network to peer handsets, which use it to determine a reference location. The invention also discloses hierarchical groups of mobile devices (including sub-groups and priorities) for information sharing, with profile- and trust-based control over which requests are answered and whose data is accepted.

The preferred embodiment is a cellular telephone with an integrated GPS receiver. A handset in a favorable signal environment demodulates GPS navigation messages (ephemeris, almanac, clock corrections), stores them, and computes its own position. A second handset needing position assistance sends a query (via the radio base stations and core network) either to a specific

time-of-week (TOW), approximate position, ionospheric/differential

corrections, and satellite almanac. The requesting handset may compile data from multiple peers and sort it (e.g., by data age).

Additional disclosed aspects:

Hierarchical groups. Groups may have one or more levels and sub-groups; each sub-group may be assigned a priority. A phone needing assistance queries the highest-priority sub-group first, then lower-priority ones. Requests may be classified by hierarchy level, and a handset may choose to respond or not based on profile and classification. Responses may be accepted only from certain group members.

Group formation. Groups form ad-hoc, with or without user initiation, or automatically based on user profiles (e.g., profile indicates joining a group for a class of information-sharing service when one is found). A group leader/host may maintain membership and enforce privacy constraints; a server within the cellular or backbone network may alternatively maintain group membership and status.

Trust model. Because peer-supplied assistance data is not guaranteed valid, a client may assign trust levels to groups (e.g., inner circle = family, outer circle = friends and associates, world = everyone else) and accept assistance accordingly, falling back to a secure fixed server if no trusted peer source meets criteria.

Coordinated sharing techniques. (a) Regularly scheduled collection: group members take turns downloading ephemeris (valid 2-4 hours) directly from satellites and distributing it to peers via cheap messaging, staggering schedules so one member refreshes every 2 hours. (b) Supply from a peer with good signal-to-noise ratio to a peer in a disadvantaged environment (e.g., indoor) via the cellular infrastructure. (c) Relaying data from a network assistance area to a handset travelling into an unassisted area. (d) Cost reduction by sharing a single network-assisted download among the group.

Other data. Exchange of phone numbers, e-mail addresses, database synchronization, games, music files. Communication over SMS, WAP, or IP. Data may also be exchanged during an ongoing voice call.

Privacy and performance profiles. Users control privacy of position and aiding information (globally or per user/group). Profiles may encode desired standby/talk time trade-offs, e.g., limiting assistance requests to one every 30 minutes or on an as-needed basis.

The patent discusses the requirement for a cellular phone to know its cell, location-sensitive applications (targeted advertising, travel directions, enhanced 911), and the desirability of providing GPS assistance data to handsets to improve time-to-first-fix and sensitivity. It cites prior network-based assistance approaches: US 5,365,450 (Schuchman et al.), US 5,418,538 (Lau), US 5,883,594 (Lau). It identifies the problem that a GPS-equipped phone may sit in a cell lacking GPS assistance data, and the desire for an alternate source.

Claim 1 (independent, apparatus): a mobile terminal comprising a GPS receiver to receive GPS data, a cellular transceiver to communicate with a wireless communications network, wherein the terminal generates GPS assistance data from the received GPS data and transmits the GPS assistance data to a remote mobile terminal via the wireless communications network.

Claim 15 (independent, apparatus): a mobile terminal with a cellular transceiver, wherein the terminal receives GPS assistance data via the wireless network generated by a remote mobile terminal from GPS data the remote received, and determines a reference location from that assistance data.

Claim 23 (independent, method): receiving GPS data from an external source, generating GPS assistance data, determining whether to transmit the assistance data to a remote mobile terminal, and transmitting it based on that determination.

Claim 37 (independent, method): receiving GPS assistance data from a remote mobile terminal, determining whether to trust that data as valid, and determining a reference location based on it.

Dependent features: group membership (hierarchical, sub-groups, priorities, ad-hoc, geographic proximity, profile-defined membership); trust decisions based on hierarchy level/sub-group/priority; transmission responsive to request or automatic; periodic retrieval and distribution; profiles for transmission decisions.

Proof-of-concept status: Not established from the supplied patent PDF alone.
Product commercialization evidence: Not established from the supplied patent PDF alone.

Production or sales evidence: Not established from the supplied patent PDF alone.

Development stage for this record: Patent-disclosed implementation; engineering maturity and commercial readiness require separate evidence.

104(b).

Government rights: none disclosed in the record.

Patent term/status: the supplied PDF confirms grant on December 12, 2006, but current enforceability and expiration require a separate current-status check; this evaluation is not an FTO opinion.

Public disclosure before the January 23, 2001 filing date: **NOT ESTABLISHED.**

No inventor statement or complete disclosure history was supplied.

Patent publication dates established from the PDF: US 2002/0098849 A1 on July 25, 2002; US 7,149,534 B2 on December 12, 2006. These are post-filing disclosures.

Sale offer / commercial offer dates: **NOT ESTABLISHED.**

Required follow-up: obtain an inventor or rights-holder statement covering talks, publications, demonstrations, offers for sale, grants, and social-media disclosures, each with a date. This intake gap remains in the Operational Audit and must not be silently treated as "none."

This is a public-patent test run, not an inventor interview. Patent-record facts are separated from unestablished proof-of-concept, disclosure-history, commercialization, and current legal-status propositions. Patentability and regulatory statements in any downstream report are preliminary and not legal advice.